SpeechLP

SpeechLP®

Research Foundation & Logic Model

ESSA Tier 4 “Demonstrates a Rationale” Evidence Packet

RESEARCH-BASED
DESIGN
ESSA IV
Logic model & research base

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RESEARCH-BASED

DESIGN

ESSA IV

Logic model & research base

DEMONSTRATES A RATIONALE

ESSA Tier 4 Evidence Statement for SpeechLP

SpeechLP

This packet documents how SpeechLP's design is grounded in published research on speech sound development and intervention, describes the program's features and intended outcomes in a logic model, and outlines a planned study of its effects. Together these components address the Every Student Succeeds Act (ESSA) criteria for Tier 4, "**Demonstrates a Rationale**."

TO COMPLETE — Independent review. The OgStar packet includes a certificate signed by an independent research organization (LXD Research). SpeechLP does not have one yet, so this page is a self-attestation. Once a third-party reviewer (e.g., LXD Research, Digital Promise, ISTE, or a university partner) has evaluated this packet, replace this box with their recognition statement, reviewer name/title, signature, and review date. Do not add a reviewer seal or signature until the review is actually issued.

Reviewer name, credentials Organization	**DATE**

Understanding ESSA Evidence

SpeechLP

Educators search for high-quality, evidence-based interventions to strengthen grant applications, to support comprehensive and targeted schools, and to implement new programming. Evidence requirements under the Every Student Succeeds Act (ESSA) are designed to help states, districts, and schools identify programs, practices, products, and policies that work across various populations.

Educational programs document their evidence of design, effectiveness, and impact in order to be eligible for federal funding. ESSA defines four levels of evidence: strong (Tier 1), moderate (Tier 2), and promising (Tier 3) evidence of effectiveness, or a program that demonstrates a rationale to be effective (Tier 4). The U.S. Department of Education's non-regulatory guidance describes Tier 4 as an activity with a well-specified logic model informed by research or evaluation, accompanied by ongoing efforts to examine its effects (U.S. Department of Education, 2016).

This packet addresses the requirements for Tier 4:

- ✓ Documentation of how the product's design relates to intended outcomes, with corresponding published academic research
- ✓ A logic model describing the product's resources, activities, outputs, and outcomes
- ✓ A study of the product's effects is planned (see "Planned Evaluation")
- ★ A third-party research organization reviews the documentation for ESSA validation [pending]

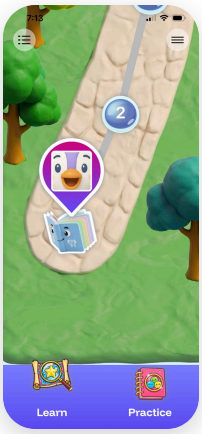


When speech-language tools are built on what is known about how children acquire and practice speech sounds, families, teachers, and SLPs can extend high-quality practice well beyond the therapy room. This packet maps each SpeechLP feature to that research and lays out how its effects will be measured.

TO COMPLETE: Replace with a quote from the independent reviewer once the review is issued.

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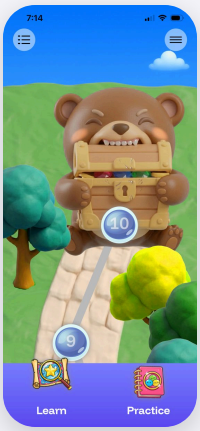
SpeechLP Supports Speech Sound Development

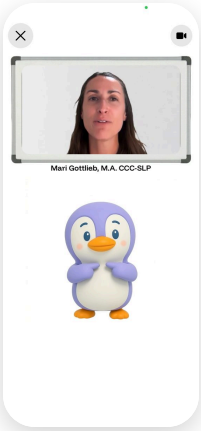


SpeechLP is a mobile app that helps children practice clear speech sounds between, or in addition to, sessions with a speech-language pathologist. A brief 24-word screener identifies which sounds are tricky for a child across word positions. The app then builds a guided learning path of short SLP-led video lessons and playful practice games drawn from a library of more than 1,000 target words, and reports progress to parents, teachers, and SLPs.

Screen, Then Practice What Matters

Children begin with a 24-word screener that samples speech sounds across initial, medial, and final word positions. Results determine which sounds and positions the child practices, so time is spent on the sounds that need attention rather than on sounds already mastered.





Taught by Licensed SLPs, Practiced Through Play

Each step of the learning path opens with a short video lesson from a certified speech-language pathologist (for example, Mari Gottlieb, M.A., CCC-SLP) who models how to produce the target sound. Children then practice in games such as *Beary Hungry*, *Baby Talk*, *Chicken Little*, *Pilot Penguin*, and *Sentence*, moving from single words to full sentences.

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User Voices

From speech-language pathologists to classroom teachers and parents, the people who support children's communication are the best judges of whether a practice tool fits their day. This section shares their experiences with SpeechLP.

TO COMPLETE — Real testimonials only. Replace each placeholder below with a genuine quote, collected with written permission to publish. Use first name and role (e.g., "Jessica, SLP, Ohio"). Aim for a mix: a school SLP, a classroom teacher, a special education director, and a parent.

S

[Quote from a school-based SLP about using SpeechLP for home practice or to support caseload.]
Name
Speech-Language Pathologist, State

[Quote from a classroom or special education teacher about independent practice in class.]
Name
Teacher, State

T

D

[Quote from a special education or related-services director about implementation or reporting.]
Name
Director of Special Education, District

[Quote from a parent about their child's motivation, confidence, or progress.]
Name
SpeechLP parent

P

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SpeechLP® Foundational Research Summary

Introduction

Speech sound disorders are among the most common communication difficulties of childhood. In a large population-based sample, Shriberg, Tomblin, and McSweeny (1999) found that about 3.8% of 6-year-olds had a speech delay, and speech sound disorders make up a large share of school SLP caseloads (ASHA, n.d.). Children with speech sound disorders are at increased risk for later difficulties with reading and spelling, particularly when difficulties persist or co-occur with language impairment (Anthony et al., 2011), and residual speech errors in school-aged children have been linked to social, emotional, and academic impacts (Hitchcock et al., 2015).

Effective speech intervention depends heavily on practice. Children need many accurate productions of a target sound, distributed over time, to make lasting gains (Maas et al., 2008; Williams, 2012). Yet the amount of practice a child receives in typical school-based service—often short, small-group sessions once or twice a week—is limited (Sugden et al., 2018).

SpeechLP is a mobile app that extends speech sound practice beyond the therapy session. It combines a brief screener, SLP-led video lessons, a large library of target words, and game-based practice that moves from words to sentences, with progress reports for parents, teachers, and SLPs. SpeechLP is designed to supplement, not replace, assessment and intervention by a licensed speech-language pathologist.

Research on SpeechLP's Key Differentiating Features

1. Targeted Screening Across Sounds and Word Positions

Children acquire English consonants in a broadly predictable developmental sequence, and accuracy for a given sound often differs depending on its position in a word (Smit et al., 1990; Crowe & McLeod, 2020). Identifying which sounds and positions are in error is the starting point for selecting intervention targets (McLeod & Baker, 2017).

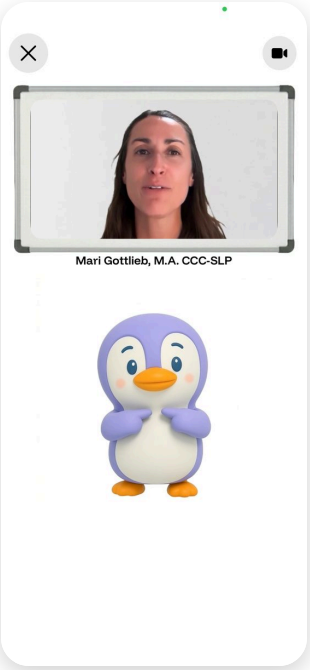
SpeechLP's 24-word screener samples speech sounds across initial, medial, and final positions. Results personalize the learning path so each child practices the specific sounds and positions that are difficult for them. Screener results are shared in a report that parents can bring to their child's SLP or teacher, supporting earlier conversations about whether a full evaluation is warranted. The screener is not a diagnostic assessment and does not replace an evaluation by a licensed SLP.

TO COMPLETE: Add a screenshot of the screener and screener report (none were available in the app-store screenshot set), and briefly describe how the 24 words were selected (e.g., reviewed by which SLPs, aligned to which developmental norms).

2. Expert Modeling Through SLP-Led Video Lessons

Learning a new speech movement benefits from clear instruction and a model to imitate. Motor learning research applied to speech emphasizes pre-practice—explaining and demonstrating how a sound is made—before a child begins repeated practice (Maas et al., 2008). Video modeling is also a well-established, effective instructional technique for children (Bellini & Akullian, 2007).

Each step of the SpeechLP learning path opens with a brief video from a certified speech-language pathologist who explains and demonstrates the target sound. Because lessons are recorded, every child receives a consistent, expert model, and families without easy access to an SLP still see correct placement and production.



Each lesson begins with a certified SLP modeling the target sound on video, followed by a character guide who leads the child into practice.

3. High-Dose Practice From a Library of 1,000+ Words

The number of practice trials matters. Children with speech sound disorders who produce more targets per session tend to make greater gains (Edeal & Gildersleeve-Neumann, 2011), and systematic reviews identify treatment intensity as a key ingredient of effective speech intervention (Kaipa & Peterson, 2016). Motor learning principles further suggest that practicing a sound across many different words supports generalization (Maas et al., 2008).

SpeechLP draws from a library of more than 1,000 words, organized by sound and word position, so children can produce many different exemplars of their target sound rather than repeating the same small set. Practice progresses in complexity—from single words in games like *Pilot Penguin* to full sentences in the *Sentence* game—mirroring the traditional hierarchy that moves from words toward connected speech (McLeod & Baker, 2017).



Children choose their target sounds, then practice them in words (Pilot Penguin) and in sentences (Sentence), building toward use in everyday speech.

4. Motivating, Purposeful Game-Based Practice

Repetitive drill can be hard for young children to sustain. A review of empirical studies found that gamification generally produces positive effects on engagement and motivation, with outcomes depending on the design and context (Hamari et al., 2014). Studies of tablet-based speech practice games for children have reported high engagement and production gains comparable to traditional paper-based practice (Hair et al., 2021), and computer-based approaches are a growing part of speech sound intervention (Furlong et al., 2017).

In SpeechLP, every game is built around speaking. In *Beary Hungry* the child feeds the bear by saying the word; in *Baby Talk* and *Chicken Little* the child must "say it right to make it pass." Children can choose which sounds to play with, progress along a visual map with numbered levels and rewards, and celebrate success with animated characters. These elements are designed to increase the number of productions a child is willing to attempt in a session.



Every game requires the child to produce their target sounds to advance.

TO VERIFY: Describe exactly how the app judges a production (automatic speech recognition, adult/parent judgment, or self-rating). If automatic, add a sentence on accuracy and cite McKechnie et al. (2018), already in the reference list.

5. Connecting Home, School, and the SLP Through Progress Reporting

Parents and caregivers can play a meaningful role in speech intervention. A review of parent involvement in intervention for speech sound disorders found that parents are frequently involved in delivering home practice and that such involvement can support outcomes (Sugden et al., 2016). In related areas of communication, parent-implemented interventions have shown positive effects on children's language skills (Roberts & Kaiser, 2011).

SpeechLP produces reports designed to be read by parents, teachers, and SLPs. Reports summarize screener results and the child's practice and progress over time, giving the SLP information about home practice that would otherwise be unknown, and giving families clear, concrete next steps. Because the app is guided by SLP lessons, parents and teachers do not need clinical training to support practice.

TO COMPLETE: Add a screenshot of a progress report and list what it includes (e.g., sounds practiced, number of productions, accuracy by position, time spent).

Planned Evaluation

SpeechLP is committed to building evidence of its effectiveness beyond Tier 4. The planned study below is designed so that results can support a future Tier 3 (promising) or Tier 2 (moderate) claim.

Research question	Do children who use SpeechLP in addition to their usual services show greater gains in speech sound accuracy than similar children receiving usual services alone?
Design	Quasi-experimental (matched comparison) or randomized waitlist-control design across partner schools.
Participants	Students in grades [K–3] with speech sound targets on their IEP or identified through screening; target sample [n].
Duration & dosage	[e.g., 10–12 weeks; 3–5 sessions per week, 10–15 minutes each]
Outcome measures	Percent consonants correct or percent target sound accuracy on untrained probe words, pre/post; standardized articulation measure [name]; app usage data; parent/teacher surveys.
Research partner	[Independent evaluator or university partner, and timeline]

Districts and schools interested in participating in a SpeechLP research study are invited to contact [contact email].

Initial Evidence of Effectiveness

TO COMPLETE — only if real data exist. The OgStar packet reports results from a small randomized trial. If SpeechLP has any real outcome data (pilot study, pre/post screener changes across users, usage and completion rates, SLP-reported progress), summarize them here with sample size and how they were collected. If none exist yet, delete this subsection; the Planned Evaluation above is sufficient for Tier 4.

[Optional chart of real pilot results, e.g., pre/post target sound accuracy]

Implementation and Instructional Support

SpeechLP runs on iPhone and iPad, so children can practice at home, in the classroom, or in the therapy room. Families can begin with the screener in minutes, and the guided learning path tells the child what to do next, minimizing setup for adults.

TO COMPLETE: Describe school onboarding (training for SLPs and teachers, support channels and hours), rostering/single sign-on, how an SLP assigns or monitors students, data privacy compliance (COPPA, FERPA, state student privacy agreements), and device requirements.

When educators use data from digital practice to guide instruction, they can better target what each child needs next. SpeechLP's reports are designed to give SLPs visibility into home practice so that session time can focus on new skills, and so that families and teachers can reinforce the same targets.

Conclusion

SpeechLP brings together several research-supported ingredients of speech sound intervention —targeted selection of sounds and word positions, expert modeling, high-dose practice across many words, a progression from words to sentences, motivating game-based practice, and family involvement—in a tool children can use between SLP sessions. Its logic model and planned evaluation provide a clear rationale for its expected impact and a path to building stronger evidence of effectiveness.

SpeechLP Logic Model

PROBLEM STATEMENT

Speech sound disorders are among the most common communication difficulties of childhood, and untreated or persistent errors can affect a child's intelligibility, literacy, confidence, and participation in school. Progress depends on frequent, accurate practice, but children typically see an SLP for limited time each week, SLP caseloads are large, and many families lack access to services or do not know how to practice at home. Practice materials are often repetitive and fail to hold children's attention. SpeechLP helps close this gap with an engaging, SLP-guided practice app that identifies each child's tricky sounds and delivers personalized, high-dose practice at home or school, with progress reports that keep parents, teachers, and SLPs connected.

RESOURCES

- 24-word screener sampling sounds across initial, medial, and final positions
- Short video lessons led by certified speech-language pathologists
- Library of 1,000+ target words organized by sound and position
- Speech practice games (e.g., Beary Hungry, Baby Talk, Chicken Little, Pilot Penguin, Sentence)
- Guided learning path with levels and rewards
- Progress reports for parents, teachers, and SLPs
- iPhone/iPad app with microphone

STRATEGIES AND ACTIVITIES

- Child completes the screener; results personalize the learning path to target sounds and positions.
- Child watches an SLP model how to produce the target sound before practicing.
- Child produces many different target words in short, game-based sessions.
- Practice progresses from words to sentences.
- Child chooses sounds to play with and advances along a map, earning rewards.
- Parents, teachers, and SLPs review reports and reinforce the same targets.
- SLPs use reports on home practice to plan sessions.

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OUTPUTS

- Each child's tricky sounds and word positions are identified.
- Children view expert SLP models of each target sound.
- Children complete a high number of target productions per session across many different words.
- Children practice targets in words and in sentences.
- Children complete practice sessions regularly at home and/or school.
- Parents, teachers, and SLPs receive regular progress reports.

SHORT-TERM AND INTERMEDIATE OUTCOMES

Improved accuracy of target sounds in words, across initial, medial, and final positions. Increased awareness of how target sounds are produced. Greater willingness to practice and increased practice frequency. Earlier identification and referral of children who may need an SLP evaluation.

As children progress through SpeechLP, they will show improved accuracy of target sounds in sentences and conversation, generalization to untrained words, improved overall intelligibility, increased confidence when speaking, stronger home–school–SLP collaboration, and more efficient use of SLP session time.

LONG-TERM OUTCOMES AND IMPACTS

- Age-appropriate speech sound production and intelligibility
- Fewer residual speech errors persisting into later grades
- Shorter time to dismissal from speech services
- Reduced risk of related literacy difficulties
- Greater confidence and participation in classroom and social communication
- Increased family capacity to support communication development

ASSUMPTIONS

- An adult (parent, teacher, or SLP) is available to support the child as needed.
- Access to an iPhone or iPad with a working microphone and a reasonably quiet practice space.
- Access to the internet.
- Children with suspected speech sound disorders are also evaluated and served by a licensed SLP where available.

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SpeechLP

SpeechLP helps children practice clear speech with SLP-led video lessons, a 1,000+ word library, and games they'll actually want to play.

For additional information about SpeechLP visit:
[website URL]

Research partnership inquiries: [contact email]